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1. Write a program using 8085 Microprocessor to add two 8-bit number.

A) APPARATUS REQUIRED

- a) 8085 kit, instruction sheet
- b) Power supply.

B) ~~Alg~~ ALGORITHM:

- a) Initialize the HL pair with address of first number.
- b) Load the first number in accumulator.
- c) Move to next memory location to get second number
- d) Add the content of accumulator with second number
- e) Store the result at specified location
- f) END the program

C) PROGRAM:

```
LXI H, 2009H
MOV A, M
INX H
ADD M
INX H
MOV M, A
HLT
```

D) RESULT

- The two numbers stored at 2009H and 2000AH are added and result was stored at 200BH.

E) CONCLUSION

Thus, the addition of two 8-bit numbers was added using assembly program in 8085 μ p.

2. WRITE A PROGRAM USING 8085 TO SUBTRACT TWO 8-BIT NUMBERS.

A) APPARATUS REQUIRED

- a) 8085 kit, instruction sheet
- Power supply

B) ALGORITHM

- a) Initialize HL pair with the address of first 8-bit number
- b) Load the first number into the accumulator.
- c) Move to the next memory location to get second number
- d) Subtract the second number from Accumulator.
- e) Store at specified location.
- f) End the program.

C) PROGRAM

LXI H, 2000H

MOV A, M

INX H

SUB M

INX H

MOV M, A

HLT

D) RESULT

The two 8-bit number stored in location 2009H and 200AH were subtracted and stored in 200BH

E) CONCLUSION

Thus, the subtraction of two 8-bit number were successfully performed.

3. WAP USING 8085 TO MULTIPLY TWO 8-BIT NUMBER.

A) APPRATUS REQUIRED

- a) 8085 Kit, instruction sheet
- b) Power supply

B) ALGORITHM

- a) Initialize the HL register pair with address of first number.
- b) Load the first number to accumulator.
- c) Move to next memory location
- d) Initialize another register with second number (B register)
- e) Clear the register where result will accumulate
- f) Add the another register to accumulator repeatedly i.e. equal to value of second number.
- g) Store the final result to specified location
- h) End the program

C) PROGRAM

```
LXI H, 2000H
```

```
MOV A, M
```

```
INXH
```

```
MOV B, M
```

```
MOV C, M
```

```
DCR B
```

```
MOV C, A
```

```
LOOP: ADD C
```

```
    DCR B
```

```
    JNZ LOOP
```

```
INXH
```

```
MOV MA
```

```
HLT
```

D) RESULT

The multiplication of two 8-bit number were stored at 200BH

E) CONCLUSION

The multiplication of two 8-bit number was performed.

4) WAP TO DIVIDE TWO 8-BIT NUMBER USING 8085.

A) APPARATUS REQUIRED

- a) 8085 Kit, instruction sheet
- b) Power supply

B) ALGORITHM

- a) Initialize HL pair with address of first number address.
- b) Load the dividend into accumulator
- c) Move to next memory location
- d) Initialize a register with divisor
- e) Clear register for storing the quotient.
- f) Subtract the divisor with dividend until result is less than divisor
- g) Increment the quotient register each time subtraction occurs.
- h) Store the quotient and remainder in specified location.
- i) End the program.

C) PROGRAM

LXI H, 2000H

MOV A, M

INX H

MOV B, M

MOV C, M

MVI D, 00H

LOOP:

SUB B

INR D

CMP B

JNC LOOP

INX H

MOV B, C

INX H

MOV B, A

HLT

D) RESULT

Thus, the division between two 8-bit num was done and quotient and remainder was stored

E) CONCLUSION

Thus division between two 8-bit num, was performed in

5 WAP using 8085 TO ADD THE ELEMENT OF AN ARRAY OF SIZE 5.

A) APPRATUS REQUIRED

- 8085 kit, instruction sheet
- Power supply.

B) ALGORITHM

- a) Initialize HL pair with starting address of array
- b) Clear the accumulator.
- c) Loop through the array element
 - Load current element
 - Add to accumulator
 - Move to next element
- d) Store Final result to specific address
- e) End the program

C) PROGRAM

```
LXI H, 2009H
MVI C, 05H
MVI A, 00H
LOOP: ADDM
      INX H
      DCR C
      JNZ LOOP
      MOV D, A
      INX H
      MOV M, D
      HLT
```

D) RESULT

The sum of 5 element of array stored in memory and stored at specific location

E) CONCLUSION

Thus, sum of all element of array was calculated in 8085.

6) WAP TO FIND THE SMALLEST NUMBER FROM ARRAY OF ELEMENTS ON 8085

A) APPARATUS REQUIRED

- = 8085 kit, instruction sheet
- = Power supply

B) ALGORITHM

- Initialize HL pair with address of first number
- Load first number of Accumulator as minimum
- Loop through the memory, load the number, check which is smaller, update and move to next element
- Store the smallest number to specified memory
- End the program.

C) PROGRAM

```
LXI H, 2000H
MOV A, M
MUI C, 0AH
INXH
LOOP: CMP M
JC SKIP
MOV A, M
SKIP: INXH
DCRL
JNZ LOOP
MOV D, A
INXH MOV
MOV M, D
HLT
```

D) RESULT

The smallest number in the array was stored in 2006H address.

E) CONCLUSION

Thus the smallest number in the array was successfully determined using assembly in 8085

7. WAP TO COUNT THE NUMBER OF ELEMENT IN ARRAY OF 8085

A) APPARATUS REQUIRED

- 8085 kit, instruction sheet
- Power supply.

B) ALGORITHM

- a) Initialize HL register with address of first element
- b) Load the first element of array.
- c) Initialize a counter register to count the number of element
- d) Loop through the array
 - Compare the memory value with FFH. If the value is FFH then that is end of array.
 - If not then loop through element until FF is found.
- e) Store the count in a memory location
- f) End the program.

C) PROGRAM

```
LXI H, 2000H
MVI C, 00H
MOV A, FFH
INX H
LOOP: CMP M
      JZ END
      INR C
      INX H
      JMP LOOP
END:
      INX H
      MOV M, C
      HLT
```

D) RESULT

The size of Array was determined by counting the element and stored

E) CONCLUSION

Thus, the size of array was successfully counted on 8085

8) MAP TO ADD TWO 16-BIT NUMBER IN 8085.

A) APPARATUS REQUIRED

- 8085 kit, instruction sheet
- Power supply

B) ALGORITHM

- a) Initialize the HL pair with address of first bit
- b) Load the 16 bit data in DE pair
- c) Move to next number location
- d) Load the 16 bit data in BC pair
- e) Move each bit in Accumulator and add in accumulator
- f) Store added to a specified location
- g) End the program

B) PROGRAM

LXI H, 2000H

MOV D, M

INX H

MOV E, M

INX H

MOV B, M

INX H

MOV C, M

MOV A, B

~~MOV~~ ADD D

MOV B, A

MOV A, C

ADD E

MOV C, A

INX H

MOV M, B

INX H

MOV M, C

HLT

C) RESULT

The sum of two 16-bit data was stored in a specific memory location

D) CONCLUSION

Thus, sum of two 16-bit num was calculated.

WAP TO SUBTRACT TWO 16-BIT NUMBER USING 8085

A) APPARATUS REQUIRED

- 8085 kit, Instruction Sheet
- Power supply

B) ALGORITHM

- Initialize the address of first number in HL pair
- Load the data in DE pair
- Move to the next number location
- Load the data in BI pair
- Sub the num in respective bit
- Store the subtraction
- End the program.

C) PROGRAM

```
LXI H, 2000H
MOV E, M
INX H
MOV D, H
INX H
MOV C, M
MOV B, INX H
MOV B, M

MOVA, E
SUB C
MOV E, A
MOV A, D
SUB B
MOV D, A
INX H
MOV M, E
INX H
MOV M, D
HLT
```

D) RESULT

The difference between the two 16-bit num was stored.

E) CONCLUSION

Thus, difference between two 16-bit number was calculated.